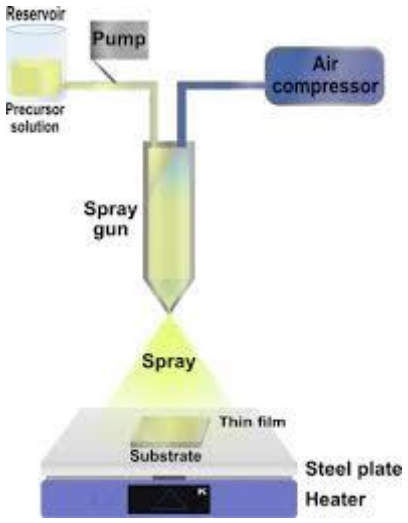
MSE1211 Crystallography II

3-Crystal Growth

Abu Ubaeda
MSE-22

1. Spray Pyrolysis

Spray pyrolysis is a solution-based thin film deposition method where a **precursor** solution is atomized and sprayed onto a heated substrate, leading to thermal decomposition of droplets and the formation of **thin films**.



Mechanism

1. **Precursor Solution Preparation:** Dissolved metal salts (e.g., zinc acetate for ZnO) in a solvent like water.

$$Zn(CH_3COO)_2 + H_2O \rightarrow ZnO + 2CH_3COOH$$

2. **Atomization:** A nozzle or ultrasonic spray breaks the solution into tiny droplets. The droplets are typically $1–10\ \mu$m in diameter.

3. **Transport via Carrier Gas:** Air, nitrogen, or inert gas may be used to transport droplets toward the hot substrate ($300–350°C$).

4. **Substrate Heating:** The substrate is kept at $300–350°C$ so that the droplets land on the surface.

5. **Thermal Decomposition:** The solute decomposes, forming a solid film and releasing byproducts:

$$Zn(CH_3COO)_2 \cdot 2H_2O \rightarrow ZnO + CO_2 + CH_4 + \text{Steam } (H_2O)$$

6. **Film Growth:** Film growth occurs with every spray cycle. Thickness is controlled by spray duration and concentration.

Abu Ubaeda
MSE_22

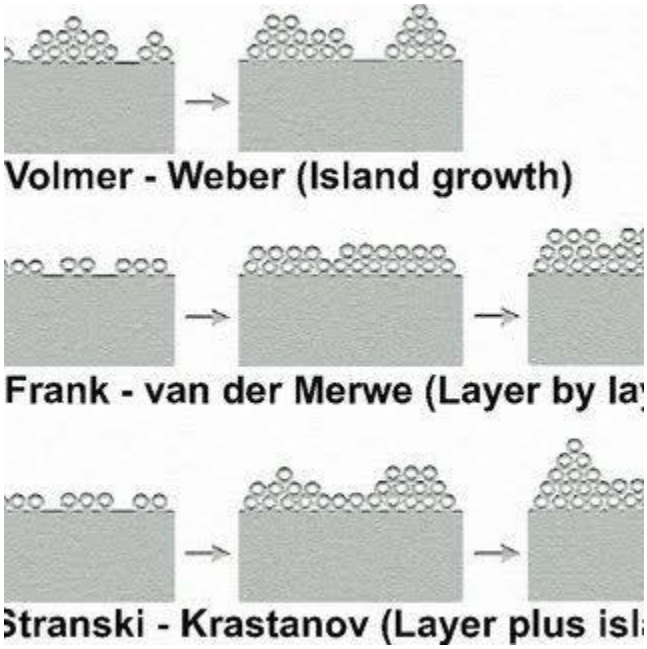
Models of Film Growth

1. **Frank–van der Merwe (Layer-by-Layer):**

   The **adsorbate–surface** and **adsorbate–adsorbate** interactions are balanced. This type of growth requires lattice matching and is hence considered an ideal growth mechanism. Each atomic layer is completed before the next one begins.

2. **Volmer–Weber (Isolated Island):**

   In this growth method, the **adsorbate–adsorbate** interaction is stronger than the **adsorbate–surface** interaction. Hence, islands are formed right away. It is also known as island growth. It is a thin film deposition process where small clusters of atoms form on the substrate and grow in three dimensions to create islands. This mode leads to a rough multi-layer structure.



3. **Stranski–Krastanov (Layer + Island Method):**

   In this growth mode, the **adsorbate–surface** interactions are stronger than **adsorbate–adsorbate** interactions. This is a thin film growth mode where initial layer-by-layer growth is followed by the formation of three-dimensional islands on top of the initial layer.

Applications of Spray Pyrolysis

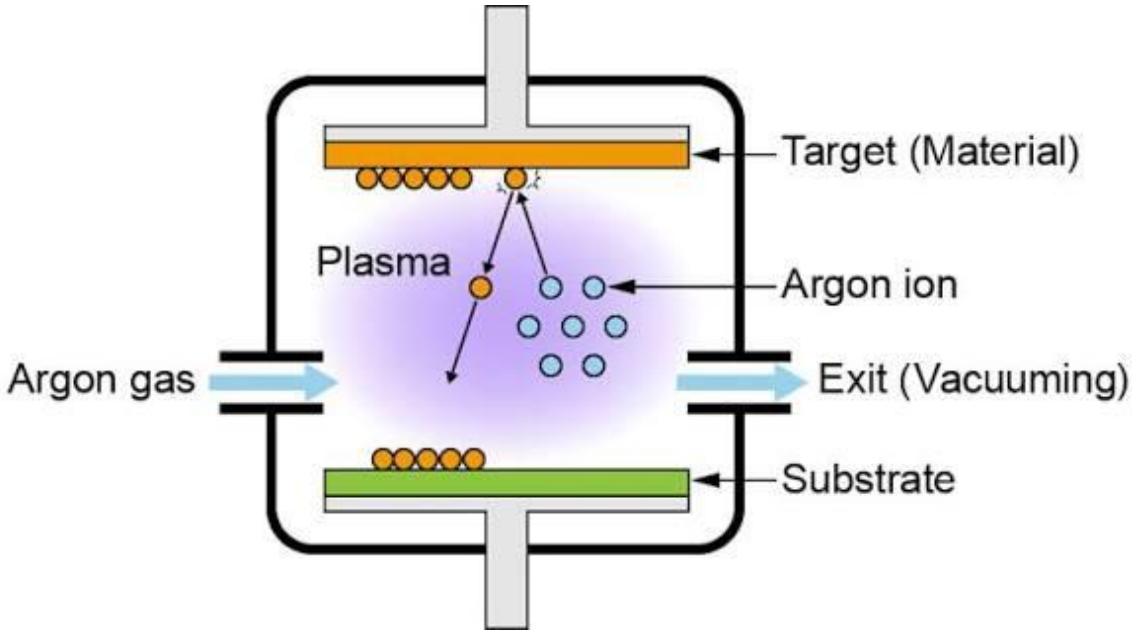
- Production of thin films
- Ceramic coating
- Solar cells
- Gas sensors
- Solid oxide fuel cells

Abu Ubaeda
MSE_22

2. Sputtering Deposition

Sputtering is a physical vapor deposition (PVD) technique in which ionized gas particles (usually Ar^+) bombard a target material, ejecting atoms that then condense on a substrate to form a thin film of about $5–250$ nm thickness.

Mechanism



1. **Vacuum Chamber:** Maintained at low pressure (10^{-5} to 10^{-6} Torr). Argon gas is introduced into the chamber.
2. **Plasma Generation:** A voltage (DC or RF) is applied between electrodes. Electrons ionize the argon gas, forming Ar^+ ions and plasma.
3. **Ion Bombardment:** Ar^+ ions are accelerated toward the negatively biased target (cathode). Collisions knock atoms **off** the surface of the target via momentum transfer.
4. **Deposition:** Ejected atoms travel in straight lines and condense on the substrate, forming a thin film.

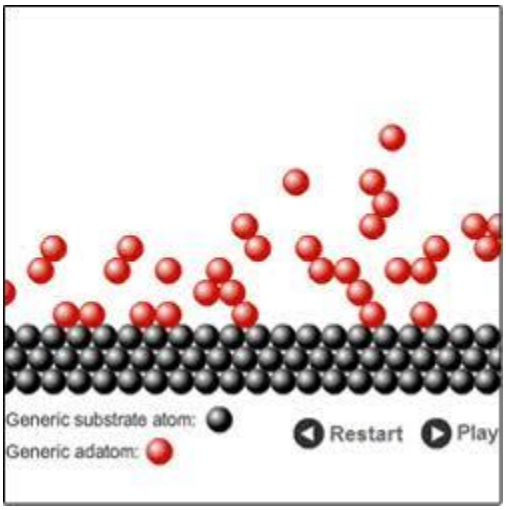
Types of Sputtering

1. **DC Sputtering:** Used for conductive targets.
2. **RF Sputtering:** Used for insulating targets (e.g., SiO_2).
3. **Magnetron Sputtering:** Magnetic field improves plasma density and efficiency.

Abu Ubaeda
MSE_22

3. Epitaxial Growth

A method of growing a crystalline layer on a substrate crystal where the grown layer follows the crystallographic orientation of the substrate is called **Epitaxial Growth**.



Types of Epitaxial Growth

There are two types of epitaxy:

1. **Homoepitaxy:** The grown layer consists of the same material as the substrate.
    - *Examples:* Si on Si, GaAs on GaAs
2. **Heteroepitaxy:** The grown layer consists of a different material than the substrate.
    - *Examples:* Ge on Si, SiC on Si

Mechanism

1. **Nucleation:** Incoming atoms (adatoms) arrive at the substrate surface.
2. **Surface Diffusion:** Adatoms migrate to energetically favorable sites.
3. **Lattice Matching:** Adatoms align with substrate atoms based on the underlying lattice structure.
4. **Layer / Island Growth:** Continuous film growth occurs monolayer-by-monolayer (Frank–van der Merwe mode) or via island formation (Volmer–Weber or Stranski–Krastanov mode).

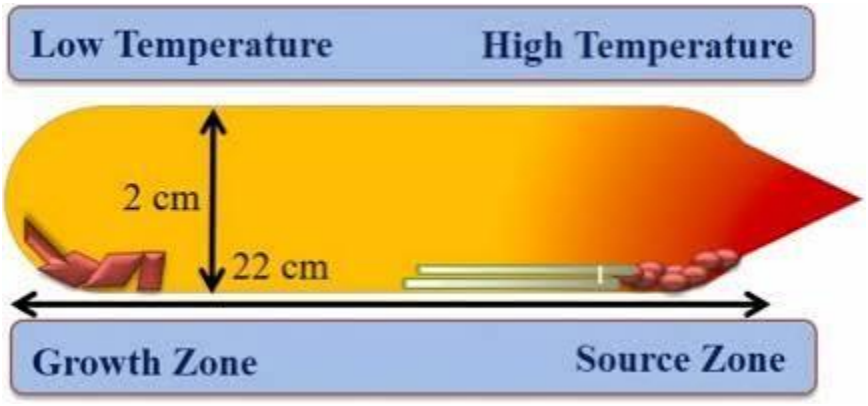
Applications

- High-speed transistors (GaAs)
- Light Emitting Diodes (LEDs) and LASER diodes (GaP, GaN)
- Solar cells
- Quantum wells and superlattices

Abu Ubaeda
MSE_22

4. Vapor Phase Transport (VPT) Single Crystal Growth

Vapor Phase Transport (VPT) is a gas-phase crystal growth method where a solid source material sublimates or reacts with a gas agent to form a vapor species, which then deposits in a cooler zone to grow a single crystal.



Mechanism

1. **Source Zone (High T):** Solid materials react with a transport gas (I_2, Cl_2, H_2, etc.) or sublimate directly to form volatile compounds.
2. **Transport:** The gas phase carries volatile species to the cooler zone, driven by a temperature gradient.
3. **Deposition Zone (Low T):** Vapor decomposes or reacts with carrier gas, undergoing reverse solid crystallization in an ordered way to form a single crystal.

Applications

- Growth of ZnO, GaN, and SiC single crystals.
- Widely used in LED substrate manufacturing.
- Compound semiconductor industries.

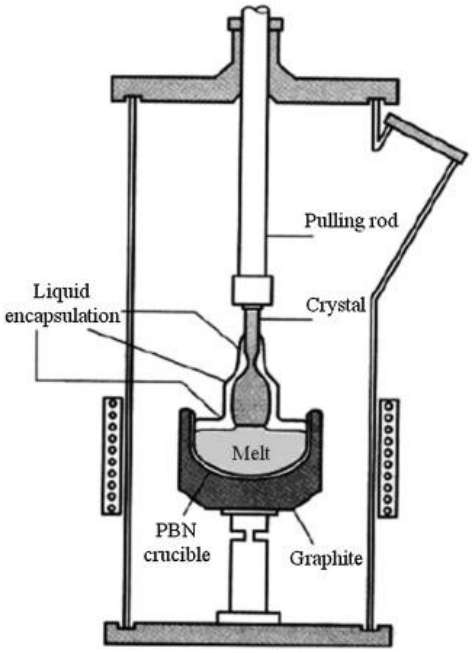
Abu Ubaeda
MSE_22

5.Czochralski Process:

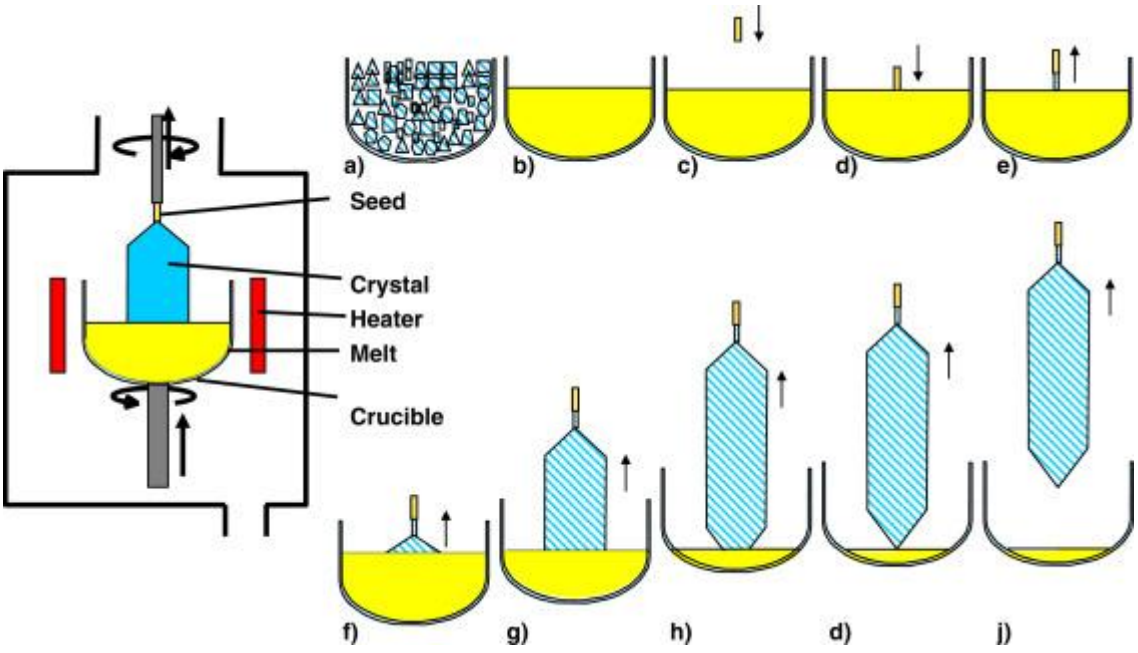
A crystal grow from a melt of the same composition by starting with a seed crystal in contact with surface. The melt whose temperature is remained slightly above its melting point.

As the seed is gradually pulled out of the melt fig (1), the melt solidifies on the surface of the seed to give a rod-shaped crystal in the same crystalline orientation as the original seed.

The melt and growing crystal usually rotated counter-clockwise during pulling.



Steps



a) Loading: Polycrystalline silicon and dopants are placed into the crucible.

Abu Ubaeda
MSE_22

b) Melting: The material is heated to high temperatures until completely melted into a liquid state.

c) Seed position: A small single-crystal seed crystal is aligned above the molten silicon.

d) Dipping: The seed crystal is lowered into the liquid melt to initiate crystal growth.

e) Necking: The seed is slowly lifted while rotating, allowing a narrow neck to form to eliminate crystal dislocations.

f) Crown formation: The pulling speed and temperature are adjusted to broaden the crystal diameter, forming the shoulder/crown.

g) Body growth: The crystal is continuously pulled upward at a constant rate and rotation to form the cylindrical main body (ingot).

h) End of body growth: Body pulling completes as the melt volume decreases significantly.

i) Tail-off (tapering): The diameter is gradually reduced to form a conical end cone, preventing thermal shock and dislocation propagation back into the ingot.

j) Separation: The completed single-crystal ingot is completely lifted out of the crucible and allowed to cool.

Application

Semiconductor Electronics

- **Integrated Circuits (ICs):** Forms the foundational silicon wafers used to manufacture microprocessors, RAM, memory chips, and analog microchips.
- **Power Electronics:** Supplies single-crystal substrates for high-voltage power devices like MOSFETs, IGBTs, diodes, and thyristors.
- **Compound Semiconductors:** Used in variant forms (e.g., Liquid Encapsulated Czochralski) to grow materials like gallium arsenide (GaAs) and germanium (Ge) for high-frequency microelectronics.

Optics & Photonics

- **Synthetic Lasers:** Produces synthetic crystals like Yttrium Aluminium Garnet (YAG) for solid-state laser systems.
- **Infrared & Optical Components:** Used to grow high-purity germanium and synthetic sapphire for IR windows, optical lenses, and specialized gemstones (e.g., synthetic alexandrite).
- **Radiation Detectors:** Supplies high-purity single crystals for gamma-ray and particle detectors used in nuclear technology and medical imaging.

Renewable Energy

- **Solar Photovoltaics:** Produces monocrystalline silicon (Mono-Si) ingots used to fabricate high-efficiency solar cells.

Advantages

- **High Crystal Perfection:** Produces high-purity single crystals with minimal structural defects and no grain boundaries, ideal for semiconductor devices.
- **Large Ingot Diameters:** Capable of growing large-diameter ingots (up to 300 mm or 450 mm), maximizing wafer yield for high-volume chip manufacturing.
- **Controlled Doping:** Allows precise uniform distribution of p-type or n-type dopants throughout the melt to achieve specific electrical properties.
- **Scalability & Maturity:** A well-established, highly automated industrial process with predictable, repeatable yields.

Disadvantages

- **Impurity Contamination:** The high-temperature molten silicon dissolves small amounts of oxygen and carbon from the quartz (SiO_2) crucible, which can affect electrical performance.
- **High Energy & Cost:** Requires continuous high temperatures (above 1414°C) for days, leading to substantial energy consumption and expensive equipment.
- **Single-Use Crucible:** The quartz crucible typically degrades or cracks during thermal cycling and must be replaced after each growth run.

- **Thermal Stress & Dislocations:** Thermal gradients between the melt and the growing ingot can induce thermal stress, leading to crystal lattice dislocations if pull rates are not strictly managed.

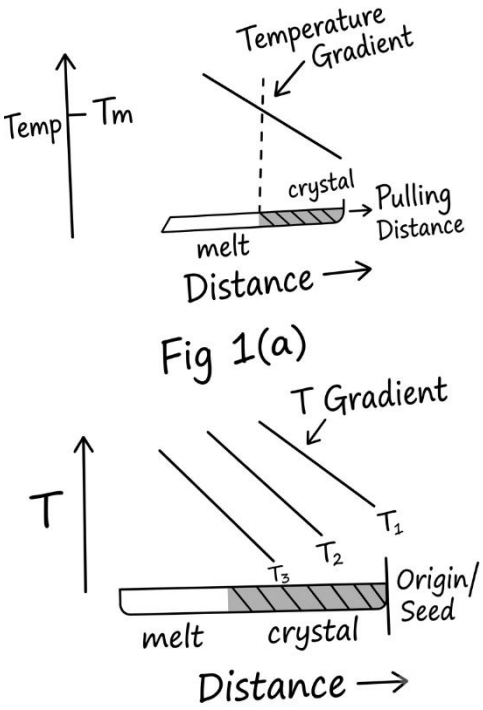
6.Bridgman–Stockbarger Method :

These methods are based on the solidification of a stoichiometric melt, where crystallization is controlled by passing the melt through a temperature gradient so that crystallization occurs at the cooler end.

This is achieved in the Stockbarger method by arranging the physical displacement of the melt container relative to a fixed temperature gradient (**Fig. 1a**).

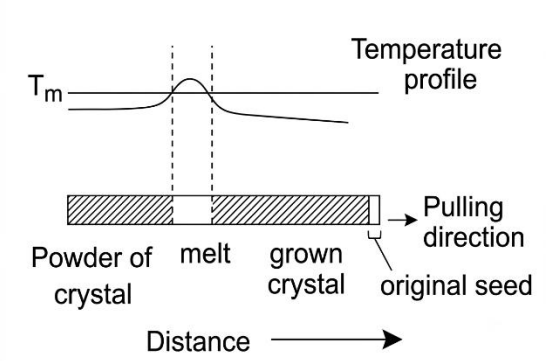
In the Bridgman method, the melt is kept inside a furnace with a fixed temperature gradient, and the furnace is gradually cooled so that solidification begins at the cooler end (**Fig. 1b**).

In both methods, it is advantageous to use a seed crystal to control crystal orientation, and protective atmospheric control may be necessary depending on the material.



7.Zone Melting / Zone Refining :

This method is closely related to the Stockbarger method, but the thermal profile through the furnace is controlled such that only a small portion of the charge is molten at any single time (**Fig.**).



Initially, the part of the charge in direct contact with the seed crystal is melted. As the boat holding the charge is pulled through the furnace, oriented solidification onto the seed occurs at the trailing edge, while more of the charge melts at the leading edge.

This is a well-known method for the purification of solids. The zone refining technique makes use of the principle that impurities are generally more soluble in the liquid phase than in the solid phase. Impurities are therefore swept along and out of the crystal by the moving molten zone. This method is widely used for crystal growth and purification of high-melting-point metals such as tungsten.